Exact Avoid-or-Compress Extrema for Dense Reachable Images: Defects Four and Five

Qichao Wang
Hebei University of Technology
Daoyu Dong
University of Electronic Science and Technology of China

October 28, 2026

Abstract

Let S be a reachable image of a strongly connected synchronizing automaton and $\varnothing \neq A \subsetneq S$. We determine the maximum shortest avoid-or-compress length over binary and arbitrary finite alphabets for $m = n - |S| = 4$ and $m = 5$. The exact values are $\min(n - |A|, 11)$ and $\min(n - |A|, 14)$, respectively. Uniform binary constructions attain both bounds. For the upper bounds, short-pair graphs and actual-image constraints reduce all parameters to finite necessary domains. Exhaustive checks exclude the domains, apart from a single relaxation eliminated by a multipartite overlap lemma. The largest calculation covers 3,206,988 domains.

1 Problem and results

All words act on the right and length counts *original alphabet letters*. A complete deterministic automaton is a triple $\mathcal{A} = (Q, \Sigma, \delta)$, where Q and Σ are finite and δ is total. It is strongly connected if every state reaches every other state, and synchronizing if $|Qw| = 1$ for some word w. Let $n = |Q|$, let $S = Qv$ be an actual reachable image with $|S| = s$, and let $\varnothing \neq A \subsetneq S$, $|A| = r$. The admissible parameters are $n \geq 2$ and $1 \leq r < s \leq n$. Put

$$\mathrm{ac}(S, A) = \min\{|w| : A \not\subseteq Sw \text{ or } |Sw| < s\}, \qquad c(S) = \min\{|w| : |Sw| < s\}.$$

Thus avoidance means omitting *at least one* protected state, and $\mathrm{ac}(S, A) \leq c(S)$. Let $\mathcal{F}(n, s, r)$ maximize $\mathrm{ac}(S, A)$ over strongly connected synchronizing automata with arbitrary finite alphabets; $\mathcal{F}_{\mathrm{bin}}$ restricts to binary alphabets. Write $m = n - s$ and $q = n - r$. The avoid-or-compress lemma [5, 6] and the Frankl–Pin compression bound [2, 4] give

$$\mathcal{F}_{\mathrm{bin}}(n, s, r) \leq \mathcal{F}(n, s, r) \leq \min\{q, \tbinom{m+2}{2}\}. \tag{1}$$

A proof of the q bound appears in Appendix A. Open Problem 4 of the survey [6] asks for tight bounds depending on the set cardinalities. The two-state image case is already illustrated there, and related fixed-state avoiding-or-compressing sharpness is discussed in [1]. Our main contribution concerns entire dense-image parameter families.

Theorem 1 (Dense-image extrema; computer-assisted upper bounds)**.** *For all admissible parameters,*

$$\boxed{\mathcal{F}_{\mathrm{bin}}(n, n-4, r) = \mathcal{F}(n, n-4, r) = \min(q, 11)}$$

and

$$\boxed{\mathcal{F}_{\mathrm{bin}}(n, n-5, r) = \mathcal{F}(n, n-5, r) = \min(q, 14).}$$

In every strongly connected synchronizing automaton, every $(n-4)$-subset has $c(S) \le 11$ when $n \ge 11$, and every $(n-5)$-subset has $c(S) \le 14$ when $n \ge 16$. These pure-compression assertions do not require reachability.

The lower bounds are established by explicit binary automata. For the upper bounds, Sections 3 and 4 present the short-pair reductions, exhaustive finite-domain coverage, and computational basis. The mathematical supplement is `PROOF.md` [7]. The universal q bound handles the parameters below the pure-compression thresholds. The codimension-three bound $C_3 \le 9$, used in both upper proofs, has a paper proof in the supplement.

Proposition 2 (Binary sharpness below the two-block threshold)**.** *Let $m \ge 4$ and $m+1 \le q \le 2m+1$. For every admissible r there is a binary strongly connected synchronizing automaton with a reachable $(n-m)$-image S and an r-set $A \subsetneq S$ such that $\mathrm{ac}(S, A) = q$.*

Proof. Appendix A supplies all three required constructions, including their reaching and reset words. The first cardinality face covers $q = m+1$; the second covers $q = m+2$. For $m+3 \le q \le 2m+1$, take $g = q-m-1$ and $M = q-1$ in Proposition 11. Then $g \ge 2$, $M \ge 2g$, $n = r+M+1$, $s = r+g+1 = n-m$, and $\mathrm{ac}(S, A) = M+1 = q$. □

2 The binary witness

Figure 1 gives a block-and-chain view of the two original letter actions for a fourteen-state instance. The diagram isolates the single merger under b and the cycle structure that makes the lower bound sharp.

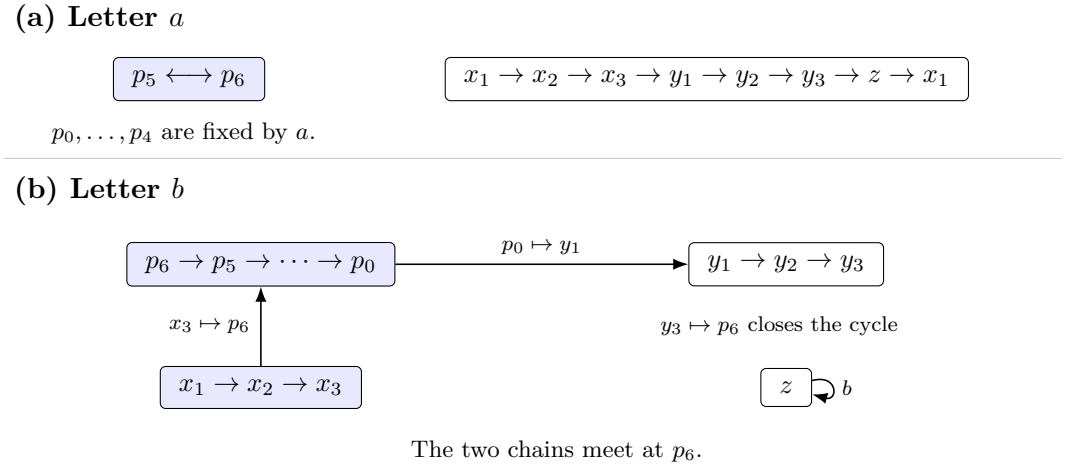


Figure 1: The original-letter actions in the binary witness $t = 3$, $p = 7$, $r = 1$ ($n = 14$). The initially occupied states are $S = P \cup X$, and $A = \{p_0\}$ is protected. Letter a is a permutation; the only nontrivial kernel of b is $\{x_3, y_3\} \mapsto p_6$. The word a^6ba^3b compresses S in 11 letters, and a^6bab^5 avoids p_0 in 13 letters.

2.1 Uniform construction and proof

Theorem 3. *Put $m = n - s \geq 3$ and $q = n - r \geq 2m + 2$. There is a binary strongly connected synchronizing automaton with reachable S attaining $\mathrm{ac}(S, A) = \min\{q, 3m - 1\}$. Consequently $\mathcal{F}_{\mathrm{bin}} = \mathcal{F} = q$ whenever $2m + 2 \leq q \leq 3m - 1$. For larger q the construction gives only a lower bound.*

Proof. Write $t = m - 1 \geq 2$ and $p = n - 2t - 1 \geq r + 3$. Take $P = \{p_0, \ldots, p_{p-1}\}$, $X = \{x_1, \ldots, x_t\}$, $Y = \{y_1, \ldots, y_t\}$ and one further state z. Letter a interchanges the last two P-states, fixes the others, and cycles $x_1, \ldots, x_t, y_1, \ldots, y_t, z$. Letter b sends p_0 to y_1, p_i to p_{i-1} for $i \geq 1$, increments indices in X, Y below t, sends both x_t, y_t to p_{p-1}, and fixes z. Put $S = P \cup X$, $A = \{p_0, \ldots, p_{r-1}\}$ and $L = 3t + 2$.

No compression before L. For $\rho_i = \min\{i, p - 2\}$ define a symmetric potential Φ on unordered pairs, zero on singleton pairs, by the following table. Indices in X, Y run from 1 to t, and indices in P from 0 to $p - 1$.

pair	potential
P–P, P–X, X–X (distinct)	L
$\{x_i, y_j\}$, $j = i$	$t - i + 1$
$\{x_i, y_j\}$, $j > i$	$2t - i + 1$
$\{x_i, y_j\}$, $j < i$	$L - j$
$\{x_i, z\}$	$2t - i + 1$
$\{y_i, y_j\}$, $i < j$	$L - j$
$\{y_i, z\}$	$2t + 1$
$\{p_i, y_j\}$	$\min\{L, \max\{L - j, 2t + 3 + \rho_i\}\}$
$\{p_i, z\}$	$\min\{L, 2t + 2 + \rho_i\}$

Substitution in the two letter actions gives

$$\Phi(E) \leq 1 + \Phi(Ea), \qquad \Phi(E) \leq 1 + \Phi(Eb).$$

Here are the boundary checks. The last-two-P interchange preserves ρ; a downward P-step decreases it by at most one. For P–Y pairs the a-boundary $y_t \to z$ decreases the second term from $2t+3+\rho$ to $2t+2+\rho$. The b-boundary $p_0 \to y_1$ with $j < t$ has source $L - j$ and target $L - j - 1$, since $j \leq t - 1$; with $j = t$ its target is at least $L - 1$. For X–Y, crossing x_t under a gives Y–Y with potential $L - j - 1$ when $j < i$; crossing under b gives P–Y of at least that potential. The kernel pair $\{x_t, y_t\}$ has potential one. Crossing y_t in a Y–Y pair under a gives Y–z of potential $2t + 1 = L - t - 1$. For X–z, the a-boundary is $\{x_t, z\} \to \{y_1, x_1\}$, with values $t + 1, t$; for Y–z, every a-image has potential $2t$. All remaining cases either retain potential, decrement an index by one, or enter a type whose potential is at least the source minus one. Clipping at L cannot increase a positive difference. Thus the inequalities hold for every index, not just tested sizes. Every distinct pair in S has potential L. Telescoping along a merging word proves its length is at least L; every compression merges an initial pair, giving the claimed lower bound.

No avoidance before q. Use singleton P-cells and t color cells, giving x_i, y_i color i and z color 1. Let T_a swap the last two P-cells and cycle the colors, and let T_b cycle $p_0, 1, \ldots, t, p_{p-1}, \ldots, p_1$. For the incidence matrix C both letters satisfy

$$D_\ell C = C T_\ell + f_z(e_1 - e_2).$$

Here D_ℓ is the original-letter transition matrix, f_z the z-column, and e_i the color row; the displayed discrepancy is a rank-one outer product. Since S has one state in every cell, induction along each original word w gives $\chi_S D_w C = \mathbf{1} + \sum_j \mu_j (e_1 - e_2) T_{w_j}$, where $\mu_j \geq 0$ counts trajectories at z

immediately before the corresponding suffix w_j. A trajectory from X needs at least $t+1$ steps to reach z; any entrance from P starts at y_1 after at least one step and then needs at least t more. Thus the first nonzero correction has position at least $t+2$. In the prescribed cell graph, color 2 needs $t-1$ letters to enter P at p_{p-1} and at least $p-r$ more to reach A. The only interchange is between adjacent last P-points and cannot shorten this descending path. A negative correction therefore reaches A only at length at least $(t+2)+(t-1+p-r) = p+2t+1-r = q$. Before then every protected coordinate has multiplicity at least one.

Attainment. After $a^{2t}b$, the missing P-point is p_{p-1}; after a further a it is p_{p-2}. This is the unique b^{p-r-1}-preimage of p_{r-1}. Hence $a^{2t}bab^{p-r-1}$ omits that protected point at length q. The $X \cup Y \cup \{z\}$ portion after $a^{2t}b$ is $\{x_2, \ldots, x_t, y_1, z\}$. After a^t it contains x_t and y_t, so $a^{2t}ba^tb$ compresses at length L. The existence of x_t in the preceding set uses $t \geq 2$. Combining both upper witnesses with the lower bounds gives $\mathrm{ac}(S, A) = \min\{q, L\}$.

Structural requirements. Put $B = P \cup Y$, $h = p+t$ and $E = B \cup \{z\}$. Letter b cycles B in the order $p_0, y_1, \ldots, y_t, p_{p-1}, \ldots, p_1$, and $Qb^t = E$. The a-cycle connects all other states to B, proving strong connectivity. On E, $d = a^{2t}b^h$ fixes P, cycles $y_1 \to y_t \to \cdots \to y_1$, and sends z to y_t. Thus $e = d^t$ fixes B and sends z to y_1. Since $Ba^{t+1} = S$, the original-letter word b^tea^{t+1} reaches S. The word a^{2t+1} fixes $X \cup Y \cup \{z\}$ and transposes the last two P-points, adjacent on the b-cycle. These words generate every permutation of B, with inverses realized by positive powers. Moreover $Ba = (B \setminus \{y_1\}) \cup \{z\}$, $Bab = (B \setminus \{y_2\}) \cup \{z\}$, and $Babe = B \setminus \{y_2\}$. Permutations route any pair in a current image into the kernel of abe; repeated compression synchronizes B. Prepending b^te synchronizes Q. For $q \leq L$, the construction and the universal bound $\mathrm{ac}(S, A) \leq q$ determine both maxima. □

3 Short-pair graphs and the upper bounds

Let P_h join distinct states if some original-letter word of length at most h merges them. Write $\tau(P_h)$ for its minimum vertex-cover size. A set survives compression through depth h exactly when it is independent in P_h. Thus

$$\exists T,\ |T| = n-m,\ c(T) > h \quad \Longleftrightarrow \quad \tau(P_h) \leq m. \tag{2}$$

This is the short-pair viewpoint on subset compression; see also the computational compression framework in [3]. Let $K_h = \{v \in Q : \deg_{P_h}(v) > 0\}$ be the nonisolated core and $J_h = Q \setminus K_h$ the isolated-state reservoir. If T is an independent $(n-m)$-set, its complement $C = Q \setminus T$ is a vertex cover of size m. As isolated vertices cover no edges, $C \cap K_h$ is also a vertex cover. Consequently

$$|(Q \setminus T) \cap J_h| \leq m - \tau(P_h). \tag{3}$$

In particular, when $\tau(P_h) = m$, every such T contains all of J_h:

$$J_h \subseteq T, \qquad Q \setminus T \subseteq K_h.$$

Thus, independently of how large n is, all m omitted states lie in the nonisolated core. Figure 2 depicts this localization for $m = 4$; its graph is schematic, not a classified core.

If $\tau(P_h) = m-1$, equation (3) allows at most one omitted isolated state in each row. The lower-cover branches retain these exceptional omissions by name, alongside the nonisolated early core; only the common, never-omitted isolated states form the unbounded reservoir. The short-image and row-rank reductions in `PROOF.md` bound this entire named support independently of n.

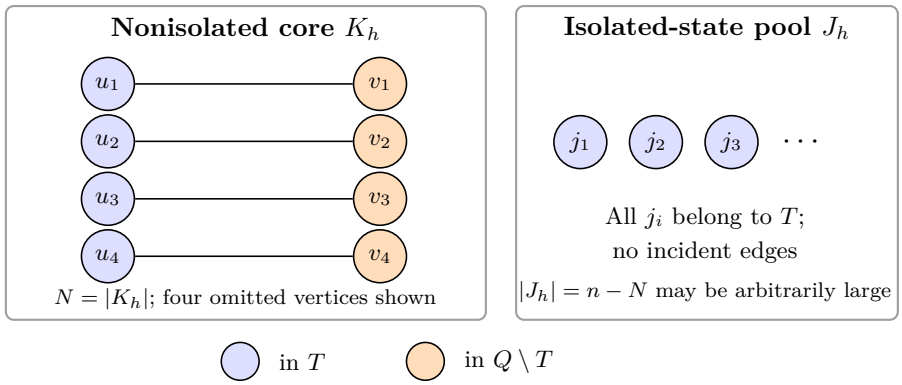


Figure 2: Nonisolated core and isolated-state reservoir in a schematic short-pair graph P_h with $\tau(P_h) = m = 4$. The four orange vertices are the omitted states $Q \setminus T$; every state in the isolated pool belongs to the independent set T. The reservoir can grow with n without changing the location of omissions. A bound on the core size N comes from the separate structural reductions, not from $\tau(P_h) = m$ alone.

Lemma 4 (Safety and union of actual short images)**.** *Suppose $|S| = n - m \geq 2$, $c(S) \geq L$, and $0 \leq d \leq L - 2$. Every Sw with $|w| \leq d$ has size $n - m$ and is independent in P_{L-1-d}. In a strongly connected automaton their union has at least $\min(n, n - m + d)$ states.*

Proof. Compression in the prefix w, or a merging suffix of length at most $L - 1 - d$, would compress S before length L. Let $U_i = \bigcup_{|w| \leq i} Sw$. If U_i is proper, strong connectivity gives an edge from U_i to its complement, whence $|U_{i+1}| \geq |U_i| + 1$. Starting at $|U_0| = n - m$ proves the union bound. □

For an early layer $G = P_h$ and a later layer P_H, the reductions retain the *complete* later maximum-independent-set family $\mathcal{D}$. Every nonisolated early-core state and every early isolated state omitted by some row is named consistently across all rows. The other isolated states lie in the common reservoir. Finite letter signatures record their possible entrances and exits. The structural support bounds and row-rank restrictions in the supplement then reduce arbitrary n to finitely many named domains.

Every original letter projects to an admissible signature. The finite domains enforce the one-step pair-distance inequalities, the existence of attaining letters, compatible row exits, and necessary strong connectivity. Where $H - h = 2$, the same original letter must also satisfy the safe-row condition for its square f^2. These constraints hold for every automaton satisfying the proposed long-compression hypothesis, so excluding all compatible finite domains proves the corresponding upper bound.

The upper-bound exclusions use the short-pair cover number to split the finite search. Figure 3 summarizes the two exact computations.

3.1 Defect four: excluding twelve steps

Assume $n \geq 11$, $|S| = n - 4$, and $c(S) \geq 12$. The known codimension-two bound $C_2 \leq 6$ and codimension-three bound $C_3 \leq 9$ imply $\tau(P_7) \geq 3$ and $\tau(P_9) \geq 4$, whereas (2) gives $\tau(P_7) \leq 4$. Here C_j denotes the maximum compression distance over all $(n - j)$-sets in the same automaton. There are exactly two branches.

1. $\tau(P_7) = 4$. The actual images of depth at most four are maximum independent sets of P_7 and cover Q by Lemma 4. The seven-layer reduction bounds the nonisolated core size by $5 \leq N \leq 8$,

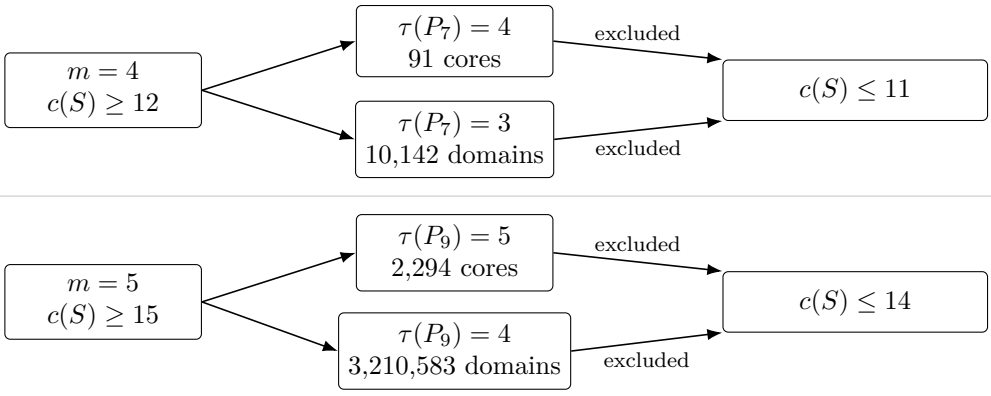


Figure 3: Case splits for the two pure-compression upper bounds. Every branch derives a finite necessary-domain contradiction; the lower defect-five branch includes 3,595 eleven-support domains and 3,206,988 smaller-support domains. Its sole surviving relaxation is excluded by Lemma 5.

its maximum degree by four, and its edge count by $7 \leq e \leq 10$, with maximum-family row rank at least five. The complete classification has 91 isomorphism classes. The complete letter and distance checks leave 152 joint distance vectors, all excluded by the necessary exit from a proper core.

2. $\tau(P_7) = 3$. Then $\tau(P_9) = 4$. The named-state reduction gives 117 early graph classes with $5 \leq N \leq 9$ and at most ten named states in total. The complete later family has three to seven rows, row rank at least three, and union omitting at most two states. The closure-compatible families, modulo core automorphisms and consistent isolated-state renaming, give exactly 10,142 necessary domains. Every domain is excluded by the common distance, quotient connectivity, and, where needed, same-letter square conditions.

The proper-core conditions apply because $n \geq 11$ exceeds the named support. The six disjoint computational ranges contain $1{,}414 + 259 + 978 + 2{,}349 + 2{,}500 + 2{,}642 = 10{,}142$ domains, with no survivors. Both complete letter-generation orders and both common-distance implementations were compared in these ranges. The two branches prove $c(S) \leq 11$; the full implication chain is in the codimension-four theorem. For $n \leq 10$, $q \leq 9$ suffices. For all n, matching binary witnesses come from Proposition 2 at $q = 5, \ldots, 9$ and Theorem 3 at $q \geq 10$.

3.2 Defect five: excluding fifteen steps

Assume $n \geq 16$, $|S| = n - 5$, and $c(S) \geq 15$. Since $C_3 \leq 9$, (2) gives $\tau(P_9) \in \{4, 5\}$. The nine-layer reduction gives the following exhaustive split.

1. $\tau(P_9) = 5$. The actual images of depth at most five are maximum independent sets and cover Q. The core has $6 \leq N \leq 10$, maximum degree at most five, at least nine edges, and maximum-family row rank at least six. Exactly 2,294 classes survive the complete graph classification. The necessary original-letter defect is at most three, using $C_4 \leq 11$: defect four would give compression within $2 + 11 < 15$. The row-exit threshold is six and fifth powers remain safe. The joint distance calculation excludes all 2,294 classes. Its 72 disjoint ranges are $[32j, 32j + 31]$ for $0 \leq j \leq 70$, and $[2272, 2293]$, so $71 \cdot 32 + 22 = 2294$.

2. $\tau(P_9) = 4$. Take $\mathcal{D}$ to be the complete maximum-independent-set family of P_{11}. It has row rank at least four and union omitting at most two states. The named support is at most

eleven. The complete early-core classification has 4,658 classes: 90 with eleven nonisolated states and 4,568 with five to ten. Each later row is either an early maximum set with one point deleted, or a nonextendable near-maximum set; each early maximum set contributes at most two later rows. The finite closure enumeration retains the identities of all omitted early isolated states.

At eleven-point early support the auxiliary-source lemma forces $\tau(P_{10}) = 4$ and exactly two early maximum sets R, T, differing in one state. Exactly one is independent in P_{10}. That same choice $R_0 \in \{R, T\}$ must be preserved into $\{R, T\}$ by *every* original letter; allowing a separate choice for each letter would discard an essential constraint. The completed coverage of the 3,595 domains is

$$\underbrace{883 + (1201 - 19)}_{\text{two omitted states: } 2065} + \underbrace{1504 + 26}_{\text{one omitted state: } 1530} = 3595.$$

The 19-domain overlap is counted once. The 1,504 single-omission domains and the final 26-domain component were checked, including both common R_0 choices in the latter.

For early support five to ten, two cores are first excluded by the entrance condition. The remaining closure enumeration and checks give

$$\underbrace{3{,}245{,}386}_{\text{raw domains}} - \underbrace{38{,}398}_{\text{static exclusions}} = \underbrace{3{,}206{,}988}_{\text{completed dynamic checks}}.$$

Exactly 3,206,987 dynamic domains are unsatisfiable. The sole satisfying *relaxation* is core 447, domain 0: the ninth-layer core is $K_{4,4}$ with parts $\{0, 1, 2, 3\}$ and $\{4, 5, 6, 7\}$, and the four later rows on that core are

$$\{0, 1, 2\}, \quad \{0, 1, 3\}, \quad \{4, 5, 6\}, \quad \{4, 5, 7\}.$$

All rows also contain the common isolated states. Figure 4 shows the two overlapping row families and the edges that the final exclusion uses.

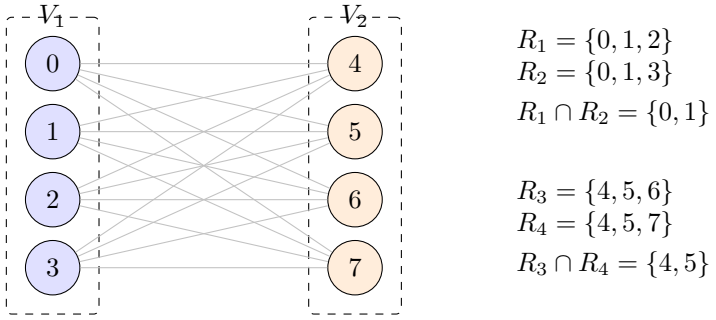


Figure 4: The sole satisfying finite relaxation: the ninth-layer core $K_{4,4}$ with all sixteen cross-part edges (gray). Each pair of overlapping triples covers its part; adjoining the common isolated set J gives safe rows in P_{10}. Lemma 5 excludes this configuration for every original alphabet.

The following paper argument excludes this remaining domain.

Lemma 5 (Multipartite overlap exclusion)**.** *Let $h \geq 1$ and $t \geq 2$. Suppose a noncomplete P_h in a synchronizing automaton is a complete multipartite graph on $V = V_1 \sqcup \cdots \sqcup V_t$ and has isolated remaining states J. Assume each V_i admits a family $\mathcal{R}_i$ of nonempty subsets whose union is V_i, whose intersection graph is connected, and for which every $R \cup J$ with $R \in \mathcal{R}_i$ is independent in P_{h+1}. Then this configuration is impossible.*

Proof. A P_h nonedge cannot be merged in one letter. Fix a letter f. It is therefore injective on each part and on J, and Jf is disjoint from Vf. Since every $(R \cup J)f$ is P_h-independent, if $Jf \subseteq J$, then $Jf = J$ and Rf is contained in some single part. Intersecting R, R' have intersecting images, so the connected intersection graph forces all of $V_i f$ into one part. Every original nonedge remains a nonedge.

Otherwise Jf meets some part V_j. Independence of $(R \cup J)f$ forces $Rf \subseteq V_j \cup J$ for every R, hence $Vf \subseteq V_j \cup J$ and $Jf \subseteq V_j \cup J$. Again every original nonedge remains a nonedge. This holds for every letter. But a synchronizing automaton with noncomplete P_h must have some letter sending a nonedge to an edge of P_h: follow the shortest merging word for a pair at distance greater than h until the distance first drops from $h+1$ to h. Contradiction. □

The two pairs of intersecting triples above separately cover the two parts of $K_{4,4}$, and their rows (with common isolates adjoined) are independent in P_{11}, hence in P_{10}. Lemma 5 with $h = 9$ excludes the only relaxation survivor. Thus the combined complete finite exclusions establish $c(S) \leq 14$. For $n \leq 15$, the original bound $q \leq 14$ suffices. Matching binary witnesses are given by Theorem 3 for $q \geq 12$ and Proposition 2: the first and second cardinality faces cover $q = 6, 7$, and the cofinal construction covers $q = 8, \ldots, 11$.

4 Computational basis and reproducibility

The upper proof consists of the short-pair reductions, the complete finite-domain checks, and the overlap exclusion in Lemma 5. Definitions and generation procedures for the finite domains are in the version-pinned supplement `PROOF.md` [7]. The principal implementation sources are:

Component	Principal source files under `scripts/problem4/`
Defect four, cover four	`compression_fourteen_cores.cpp`
Defect four, cover three	`low_cover_core_domains.cpp`; `low_cover_actions.cpp`
Defect five, cover five	`near_max_rank_cores.cpp`; `near_max_rank_actions.cpp`
Defect five, cover four	`fifteen_low_cover_domains.cpp`
Joint necessary constraints	`pair_sat_control.hpp`
Binary construction controls	`check_two_block.py`; `two_block_potential.py`

The largest finite calculation covered 3,206,988 necessary domains using 24 workers, paired letter-generation orders, and CaDiCaL 3.0.1 with joint SAT constraints (22,923.697 seconds). Of these, 3,206,987 were unsatisfiable under the checked relaxation. The remaining model is eliminated by Lemma 5. The computation used the earlier necessary-domain predicate; a subsequent strengthening to the proper-layer entrance condition is applied only in the paper exclusion.

The computation is reproducible from the source files and `docs/RUNNING.md`. Its evidential basis comprises the completed enumeration records, the stated domain-reduction arguments, program implementations and SAT decisions; per-domain UNSAT proof traces were not retained. Independent finite controls compare generation orders, small-domain enumerations, and actual automata.

5 A conjectured extension

The binary lower-bound constructions give $\mathcal{F}_{\mathrm{bin}}(n, n-m, r) \geq \min(q, 3m-1)$ for every $m \geq 4$, and Theorem 1 establishes equality for $m = 4, 5$.

Conjecture 6 (Linear defect formula)**.** *For every $m \geq 4$ and all admissible n, r,*

$$\mathcal{F}_{\mathrm{bin}}(n, n-m, r) = \mathcal{F}(n, n-m, r) = \min(n-r, 3m-1).$$

The remaining parameter region for exact avoid-or-compress extrema is $m \geq 6$ and $q \geq 3m$.

A Constructions for the smaller values of q

Theorem 7. *The following bounds hold, and every lower bound has a binary witness.*

1. $\mathcal{F}(n, r+1, r) = n - r$ *for* $n \geq 2$ *and* $1 \leq r \leq n-1$.
2. $\mathcal{F}(n, r+2, r) = n - r$ *for* $n \geq 4$ *and* $1 \leq r \leq n-3$.
3. *For* $r \geq 1$, $s - r \geq 3$ *and* $n \geq 2s - r - 1$, *one has*

$$\mathcal{F}(n, s, r) = n - r.$$

For $1 \leq r < n$, *one has* $\mathcal{F}(n, n, r) = 1$.

Thus the first two cardinality faces are determined, including their full-image boundary. For every fixed (s, r) with $s - r \geq 3$, the bound is sharp at every $n \geq 2s - r - 1$, in particular at $n = r + k(s - r - 1) + 2$ for $k \geq 2$.

We recall the upper bound with a proof, to make the argument self-contained. For a word w let $D_w(q, t) = 1$ if $qw = t$, and 0 otherwise. Then $D_{uv} = D_u D_v$. The characteristic vector χ_S is a row vector; $\chi_S D_w$ counts trajectories from S, with multiplicity, and its support is Sw.

Lemma 8 (Szykuła [5, 6])**.** *If $A \not\subseteq Sz$ for some word z, then* $\mathrm{ac}(S, A) \leq n - |A|$.

Proof. Write $r = |A|$, $s = |S|$, and let V_i be the span over $\mathbb{R}$ of $\chi_S D_w$ for $|w| \leq i$. Suppose all words of length at most $n - r$ are safe. Safety gives $|Sw| = |S|$, so $S \to Sw$ is injective: no trajectories collide, and every image state has multiplicity 1. Their rows therefore have total s and value 1 at every coordinate of A. Hence V_{n-r} lies in

$$H = \{x : x_q = s^{-1} \textstyle\sum_{t \in Q} x_t \text{ for every } q \in A\}.$$

The r equations are independent since $r < s$, so $\dim H = n - r$. Now $\dim V_0 = 1$, and $V_i = V_{i+1}$ implies invariance under every letter and stabilization forever. Therefore some equality must occur before index $n - r$, or $\dim V_{n-r} \geq n - r + 1$. Stabilization puts $\chi_S D_z$ in H, contradicting its zero coordinate in A and total s. □

In a strongly connected synchronizing automaton with $n \geq 2$, a reset word followed by a path to a state outside A gives the required z. This proves every upper bound $n - r$ in Theorem 7. If $S = Q$, some letter must be nonpermutational and immediately compresses Q, while the empty word is safe. Thus $\mathrm{ac}(Q, A) = 1$.

We will also use the following elementary fact. If t cycles $x_0, x_1, \ldots, x_{s-1}$ and e sends x_0 to x_1 while fixing the other points, then

$$e(t^{s-1}e)^{s-2} \tag{4}$$

maps this set to x_1. Indeed, after e the image is $\{x_1, \ldots, x_{s-1}\}$; each $t^{s-1}e$ deletes the last point of the remaining interval. This also covers $s = 2$.

A.1 The first two cardinality faces

Proposition 9. *Part (1) of Theorem 7 holds.*

Proof. Put $L = n - r \geq 1$ and $Q = \{0, \ldots, n-1\}$. Define

$$qa = \begin{cases} q+1 & q < L, \\ q & q \geq L, \end{cases} \qquad S = \{0, L, L+1, \ldots, n-1\}, \qquad A = S \setminus \{0\}.$$

Let b cycle S in the displayed order and send every point outside S to L. Then $Qb = S$. The a-path from 0 to L and the b-cycle prove strong connectivity.

For $0 \leq j < L$, set $S_j = A \cup \{j\}$. One has $S_j b = S_0$, and $S_j a = S_{j+1}$ for $j < L-1$; $S_{L-1}a = A$. Induction shows that every word of length less than L has image S_j with $j < L$, so is safe. The word a^L compresses S, proving $\mathrm{ac}(S, A) = L$.

On S, $c = a^L$ is the unit $0 \mapsto L$. Formula (4) with $t = b$ shows that $c(b^r c)^{r-1}$ resets S. It resets Q as well, since $Qc = A = Sc$. This proves synchronization, also when $L = 1$ or $r = 1$. □

Proposition 10. *Part (2) of Theorem 7 holds.*

Proof. Put $M = n - r \geq 3$. Take disjoint sets $A = \{p_0, \ldots, p_{r-1}\}$ and $U = \{u_1, \ldots, u_M\}$, and let $S = A \cup \{u_1, u_2\}$. Letter a fixes A, while

$$p_i b = p_{i+1} \quad (0 \leq i < r-1), \qquad p_{r-1} b = u_1.$$

For odd M, let a cycle $u_1, \ldots, u_M$ and set

$$u_j b = \begin{cases} p_0 & j \text{ odd}, \\ u_2 & j \text{ even}. \end{cases}$$

For even M (necessarily $M \geq 4$), set

$$u_1 a = u_2, \quad u_j a = u_{j+1}\ (2 \leq j < M), \quad u_M a = u_2, \qquad u_j b = \begin{cases} u_2 & j = 1 \text{ or } j \geq 3 \text{ odd}, \\ p_0 & j \text{ even}. \end{cases}$$

In both cases $Qb = S$. From p_0 the b-path reaches u_1, whose a-orbit covers U; some state of U returns to p_0 under b. Every state therefore reaches and is reached from p_0.

The two moving points of Sa^j remain distinct for all $j \geq 0$. For $0 \leq j \leq M-2$, their b-images are p_0 and u_2, so $Sa^j b = S$. At $j = M-1$, the pair is $\{u_M, u_1\}$ for odd M and $\{u_M, u_2\}$ for even M; both points map to p_0. Thus $h = a^{M-1}b$ compresses S. Every word shorter than M splits into a-runs followed by b, each run having length at most $M-2$, and a trailing a-run. All these transitions are safe. Hence $\mathrm{ac}(S, A) = M$.

Set $s = r + 2$, and $t = ab$ for odd M, $t = b$ for even M. On S, t cycles $p_0, \ldots, p_{r-1}, u_1, u_2$. If e is the unit $u_1 \mapsto u_2$ on S, then $h = et$ there. Consequently

$$h(t^{s-2}h)^{s-2} = e(t^{s-1}e)^{s-2}t$$

on S, and this is a reset by (4). Prepending b gives a reset of Q. The unit e is used only to verify the identity; the reset word uses the original letters. □

A.2 Every larger fixed gap

Proposition 11. *For $r \geq 1$, $g \geq 2$ and $M \geq 2g$, there is a binary strongly connected synchronizing automaton with*

$$n = r + M + 1, \quad |S| = s = r + g + 1, \quad |A| = r, \qquad \mathrm{ac}(S, A) = M + 1.$$

The subset S is an actual reachable image.

Proof. Take disjoint sets $P = \{p_0, \ldots, p_r\}$ and $U = \{u_1, \ldots, u_M\}$. Set $A = P \setminus \{p_r\}$ and $S = P \cup \{u_1, \ldots, u_g\}$. First suppose $M \geq 2g + 1$ and put $h = 1 + ((M - 1) \bmod g)$. Then $1 \leq h \leq g$ and $M \geq 2g + h$. Let a fix P, send u_j to u_{j+1} for $j < M$, and send u_M to u_{h+1}. Define $p_0 b = u_h$, $p_i b = p_{i-1}$ for $1 \leq i \leq r$, and

$$u_j b = \begin{cases} p_r & j \equiv h - 1 \pmod{g}, \\ u_1 & j = g,\ h \geq 2, \\ u_{h+g} & j = M, \\ u_{j+1} & \text{otherwise.} \end{cases}$$

The cases are disjoint. On S, b cycles $p_0, u_h, \ldots, u_g, u_1, \ldots, u_{h-1}, p_r, \ldots, p_1$. The segment $u_1, \ldots, u_{h-1}$ is empty for $h = 1$. Ordinary outside states follow increasing indices to the next index congruent to $h - 1$, then enter p_r. The exceptional path of length $g + 1$ is

$$u_M \longrightarrow u_{h+g} \longrightarrow \cdots \longrightarrow u_{h+2g-1} \longrightarrow p_r.$$

Its indices exist because $M \geq 2g + h$. Thus $Qb^{g+1} = S$. The a-path from u_1 covers U, and the b-cycle covers P, proving strong connectivity.

Safety. Use singleton P-cells and color cells $1, \ldots, g$, assigning u_j color $1 + ((j - h - 1) \bmod g)$. Let C be the $n \times s$ state-to-cell incidence matrix. Let T_a fix P and cycle the colors, and let T_b cycle $p_0, g, 1, \ldots, g - 1, p_r, \ldots, p_1$. For a standard state column f_q and cell row e_c, one has

$$D_a C = C T_a, \qquad D_b C = C T_b + f_{u_M}(e_g - e_1). \tag{5}$$

The last term is an outer product of an $n \times 1$ column and a $1 \times s$ row. These cells are bookkeeping devices, not a quotient automaton. For $w = \ell_1 \cdots \ell_t$, iteration gives

$$\chi_S D_w C = \mathbf{1}_s + \sum_{\substack{1 \leq i \leq t \\ \ell_i = b}} \mu_i (e_g - e_1) T_{\ell_{i+1} \cdots \ell_t}, \qquad \mu_i = (\chi_S D_{\ell_1 \cdots \ell_{i-1}})_{u_M} \geq 0. \tag{6}$$

This counts trajectories even after collisions, with products in word order. A trajectory from S needs at least $M - g$ letters to reach u_M: initial indices are at most g, every transition within U increases an index by at most one, and entrances from P have index $h \leq g$. Hence a nonzero correction has $i \geq M - g + 1$. In the cell graph, color g needs at least $g - 1$ letters to reach color $g - 1$, and color 1 needs at least g letters to reach A. The latter route traverses $1, \ldots, g - 1$ and then takes two letters b through p_r. A detour through P cannot shorten either route.

For $t \leq M - 1$, correction suffixes have length at most $g - 2$, so no positive correction reaches color $g - 1$, whose trajectory count is at most one. Suppose a word of length at most M has a first collision. For a colliding b, the distinct source states must lie in its only non-singleton fiber, all of color $g - 1$, contradicting this count bound. For a colliding a, the sources must be u_h, u_M, its only non-singleton fiber. They are not both initially present. A preceding b cannot give u_M, which

has no b-preimage. A preceding a requires sources u_{h-1}, u_{M-1} one step earlier, both of color $g-1$, again a contradiction; for $h=1$, u_h has no a-preimage at all. Thus every word of length at most M preserves s states. For $t \le M$, every correction suffix has length at most $g-1$, so no negative correction reaches A. Its singleton cells retain multiplicity at least one. Every such word is safe.

The word $a^{M-g}b^{g+1}$ has length $M+1$. Exactly its first b has a trajectory at u_M, with multiplicity one. Its negative correction is $e_1 T_b^g = e_{p_{r-1}}$, whereas its positive correction is $e_g T_b^g = e_{p_r}$. It therefore omits p_{r-1} and attains equality, also when $r=1$.

For synchronization put $c = a^{M-h}b^s$. The a-run fixes the cyclic tail $u_{h+1}, \dots, u_M$. For $j < h$, it sends u_j to u_{M-h+j}, which has the same remaining b-distance $h-j$ to p_r as u_j. For $j=h$, it sends u_h to u_M, which reaches p_0 in $g+1+r = s$ letters b. Consequently c fixes $S \setminus \{u_h\}$ and sends u_h to its b-predecessor p_0. After c, each bc extends the missing interval by one point in forward b-order. Thus $b^{g+1}c(bc)^{s-2}$ resets Q.

The boundary $M = 2g$. Let a fix P and cycle U. Define $p_0 b = u_g$, $p_i b = p_{i-1}$ and

$$u_j b = \begin{cases} p_r & j \equiv g-1 \pmod g, \\ u_{j+1} & j < M,\ j \not\equiv g-1 \pmod g, \\ u_M & j = M. \end{cases}$$

Now b cycles $B = P \cup \{u_g, \dots, u_{2g-1}\}$ and fixes u_{2g}; $Qb^{g+1} = E = B \cup \{u_{2g}\}$. Put $d = a^{g-1}b^s$. It fixes P and on E has actions

$$u_g d = u_{2g-1}, \quad u_{g+1} d = u_{2g}, \quad u_{g+j} d = u_{g+j-1}\ (2 \le j \le g-1), \quad u_{2g} d = u_{2g-1}.$$

Therefore $Eda^g = S$, proving actual reachability. The two cycles prove strong connectivity. Assigning u_j color $1 + ((j-1) \bmod g)$ gives exactly (5)–(6) again. The same delay and collision arguments apply, with a now a permutation. The word $a^g b^{g+1}$ attains avoidance: only the first negative correction has a suffix of g letters b and reaches p_{r-1}; no positive correction reaches A. Finally $e = d^g$ fixes $E \setminus \{u_g\}$ and sends u_g to u_{2g}. Each be removes one further point of B while retaining u_{2g}. Hence $b^{g+1}e(be)^{s-1}$ resets Q. □

Taking $g = s-r-1$ and $M = n-r-1$ proves part (3) of Theorem 7.

Acknowledgments

We thank Marek Szykuła for correspondence concerning Open Problem 4 and the theory of synchronizing automata.

Large language models, including OpenAI's ChatGPT, assisted with mathematical exploration, code development and checking, and manuscript drafting and editing. The authors directed and reviewed these contributions and take responsibility for the results presented here.

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